

Taxing Automation? A Case for Redistribution across Skills and Generations ^{*}

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Abstract

What are the welfare effects of labor market polarization due to automation? Can an automation tax mitigate its redistributive consequences? We develop an overlapping generations model in which individuals choose consumption, saving, and occupations over their life-cycles and accumulate occupation-specific human capital. We study technological changes that account for occupational wage growth since the early 1980s. Automation generates substantial distributional effects across skills and generations: young high-skilled individuals gain, while low-skilled entrants lose, with these differences becoming larger for later generations. We then ask whether an automation tax can redistribute these gains and losses. Compared with the status quo effective tax rate of 10%, the optimal tax on equipment income is 22%. The tax improves welfare for low-skilled individuals at the expense of high-skilled individuals, partially offsetting the distributional effects of automation. For individuals alive in 1980, aggregate welfare rises by 0.20%; low-skilled individuals gain 0.24%, while high-skilled individuals lose 0.06%. Overall, the tax provides redistribution toward those harmed by automation, but its aggregate welfare gain is relatively small.

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